

Expt 3	Performance Analysis of IIR Filters
Date	3.1 Design and Comparative Analysis of Butterworth and Chebyshev Type-I IIR Band-Pass Filters Using MATLAB

```

clc; clear; close all;

% Specifications
Fs=2000; Rp=1; Rs=40;
Wp=[300 600]/(Fs/2);
Ws=[200 700]/(Fs/2);

% Butterworth Filter
[n1,W1]=buttord(Wp,Ws,Rp,Rs);
[b1,a1]=butter(n1,W1,'bandpass');

% Chebyshev Type-I Filter
[n2,W2]=cheb1ord(Wp,Ws,Rp,Rs);
[b2,a2]=cheby1(n2,Rp,W2,'bandpass');

% Magnitude Response
figure;
freqz(b1,a1); title('Butterworth Filter');

figure;
freqz(b2,a2); title('Chebyshev Type-I Filter');

% Group Delay
figure;
grpdelay(b1,a1); hold on;
grpdelay(b2,a2);
legend('Butterworth','Chebyshev-I');

% Pole-Zero Plot
figure;
subplot(121),zplane(b1,a1),title('Butterworth')
subplot(122),zplane(b2,a2),title('Chebyshev-I')

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```
% Impulse Response
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```
figure;
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```
subplot(211),impz(b1,a1),title('Butterworth Impulse')
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```
subplot(212),impz(b2,a2),title('Chebyshev Impulse')  
% Pole Magnitude  
figure;  
subplot(121),stem(abs(roots(a1)),'filled')  
title('Butterworth Pole Magnitude')  
subplot(122),stem(abs(roots(a2)),'filled')  
title('Chebyshev Pole Magnitude')  
disp(['Butterworth Order = ',num2str(n1)])  
disp(['Chebyshev Order = ',num2str(n2)])
```





